

APPENDIX B –RF SHIELDED ENCLOSURE TEST GUIDELINE

B-1 INTRODUCTION

This document describes the procedure and methodology of performing an RF shielding effectiveness verification test on enclosures which will house GE Medical Systems (GEMS) Magnetic Resonance Imaging (MRI) equipment. MRI equipment is sensitive to RF energy from sources outside of the shielded enclosure. To ensure proper operation of the MRI equipment, the shielded enclosure must attenuate local RF signals to levels that do not cause interference.

Note

RF Shielding Performance is based on **planewave** measurements. “H” field and “E” field tests are not required, but are allowed as needed for diagnostic purposes.

B-1-1 PURPOSE OF TEST PLAN

The purpose of this test plan is to describe a series of RF shielding effectiveness tests to demonstrate compliance of an MRI shielded enclosure to the requirements of GE Medical Systems.

The test procedure described in this guideline is a modification of MIL-STD-285 and IEEE Std 299-1991. This procedure provides a thorough evaluation of the shield integrity at the upper end of the frequency range of interest showing any RF leakage which may cause imaging problems. These test guidelines ensure that the electromagnetic environment inside of the enclosure will meet the requirements of General Electric Medical Systems.

B-2 APPLICABLE DOCUMENTS

MIL-STD-285	MILITARY STANDARD ATTENUATION MEASUREMENTS FOR ENCLOSURES, ELECTROMAGNETIC SHIELDING, FOR ELECTRONIC TEST PURPOSES, METHOD OF; 25 June 1956
IEEE Std 299-1991	IEEE STANDARD FOR MEASURING THE EFFECTIVENESS OF ELECTROMAGNETIC SHIELDING ENCLOSURES; 2 July 1991

B-3 TEST SAMPLE SETUP

The shielded enclosure under test shall be set-up in a normal configuration, which consists of the following:

1. Magnet installed including all floor mounting bolts
2. RF shielded door(s)
3. Waveguide penetrations, HVAC, vents, medical gas lines, etc.
4. AC power supplied through low-pass filters
5. Patient view window, skylights, windows, hatches, etc.
6. Blank penetration panel installed, dimensionally equivalent to the GE panel and the same mounting hardware to be used with the GE penetration panel.

For safety reasons, the enclosure shall be electrically grounded during the shielding effectiveness test. Any variances from the normal configuration shall be noted in the certification report.

B-4 SHIELDING EFFECTIVENESS

This test procedure determines the worst case shielding effectiveness based on the lowest test point reading obtained. The lowest reading obtained will be the reading of the room.

B-5 MEASUREMENT PROCEDURE

To simulate the effects of external RF sources, the transmit antenna shall be located outside the enclosure on a plane parallel to the face of the enclosure wall, at a distance of 6 feet (1.8 meters) unless physically constrained to a lesser separation. The areas of least effectiveness are located by searching the inside of enclosure with the antenna connected to the spectrum analyzer.

B-5-1 Test Position

The transmitting antenna will be positioned in front of all critical areas (doors, windows, filters, penetration areas, etc.) and at a minimum of every 20 feet (6.1 meters) of wall. The receiving antenna is scanned over all panel section joints (where accessible), at floor, wall, and ceiling for a minimum of 10 feet (3.05 meters) in all directions from the location of transmitting antenna. The receiving antenna shall be at a minimum of 1 foot (0.3 meters) from the shield. For areas that are inaccessible for direct location of the transmitting antenna, the inside of that area will still be scanned using the receiving antenna with the transmitting antenna positioned in front of the adjacent wall or adjacent test position.

B-5-2 Frequency Range

The standard frequency for shielding measurements shall be 100 MHz $\pm$ 10 MHz. This allows the frequency to be adjusted slightly to avoid interference from local active transmitters and/or RF noise from other sources. Test frequency utilized shall be noted in the certification report.

B-5-3 Free Field Calibration

The incident field, i.e. "free field", is measured by the following procedure:

Position the transmit antenna parallel to the exterior wall of the enclosure at a distance of 6 feet (1.8 meters) (unless physically constrained to a lesser separation, in which case a separate reference will be established and documented at the new test distance), using horizontal polarization. The receive antenna shall be placed between the transmit antenna and 1 foot (0.3 meters) from the exterior wall of the enclosure. The receive antenna will be moved vertically and horizontally to achieve maximum signal strength. The receive antenna shall be placed no closer than 2 inches (51 mm) from the exterior wall of the enclosure and in line with the transmit antenna. The maximum received voltage at the test frequency will be recorded.

B-6 ENCLOSURE POWER REFERENCE INSULATION

To prevent personnel hazard, it is necessary for the enclosure to be properly grounded.

To minimize common mode currents, the ungrounded enclosure should be isolated from ground with a minimum of 1000 ohms of DC resistance. The isolation measurement is performed by the following procedure:

All power to the enclosure is removed. For safety reasons, an AC voltage measurement will be made to verify that no power is connected. With electrical power and intentional ground disconnected, connect the test instrument between the shielded enclosure and AC power ground. Take a reading and record the value. This test shall be made using either an isolated, current limited high voltage (>150 VDC) DC source and DMM to read drop across the limiting resistor or a megger instrument capable of reading values less than 1000 ohms. Conventional resistance meters employing test sources of 9 VDC or less shall not be used.

B-7 TEST EQUIPMENT

Test equipment shall be selected to provide measuring capabilities as described in this test guideline. The signal source, amplifier, antennas, and receiver or spectrum analyzer shall be such that the difference between the induced reference voltage and the receiver sensitivity shall be at least 6 dB greater than the required attenuation specification.

The signal source and power amplifier shall output a CW signal for a nominal test frequency of 100 MHz. Receiver or spectrum analyzer and preamplifier (if required), shall provide adequate sensitivity to permit attenuation measurements to be made to the specified limits. Dipole antennas and other miscellaneous equipment required to transmit and receive the proper RF fields shall be used.

The absolute performance calibration, of the equipment requiring calibration, shall be performed on an as needed basis in accordance with MIL-STD-45662. The calibration period shall not exceed one year. The test equipment tolerances of at least $\pm 2\%$ frequency and ± 2 dB amplitude shall be met. Equipment certifications shall be traceable to the National Institute of Standards and Technology (NIST). All equipment will be verified for proper operation between and after each series of tests by repeating the reference readings at the specified frequency(s).

B-8 DATA RECORDING AND VERIFICATION

Measurements shall be performed by qualified responsible EMC test personnel. The test must be performed in the presence of a GE representative unless other arrangements have been made by GEMS. All data collected during the course of the tests will be recorded on standardized data sheets. The data sheets will include the test location, frequency, reference level, measured enclosure level, and attenuation level.

B-9 TEST REPORT

A final certification report will be provided after the test is performed. This report will include all recorded data necessary for the evaluation of the shielded enclosure test results and will list any changes pertinent to the test set-up or the shielding effectiveness. The certification report will also include the test procedures and a list of the actual equipment used during the test.

Along with the data sheet, there will be a presentable drawing showing the shape of the enclosure, all test point locations, doors, filters, windows, and existing building walls.